

Calculating rates 1A

1 Use these words to complete the following statements:
different, first, one, quantity, written, per.

A rate compares two different things.

A rate is usually expressed by writing
how many of the first quantity correspond
to one of the second quantity.

Rates are normally written using
the word per or the symbol /.

2 Adrian types 440 words in 8 minutes.
What is his typing rate in words per minute?

Adrian types 440 words in 8 minutes.
Divide: by 8.

He types 55 words in 1 minute.
Adrian's typing rate is 55 words
per minute.

3 Maree scored 42 runs in 10 overs
of cricket. What is her scoring rate
in runs per over?

Divide: by 10.
She scored 4.2 runs in 1 over.
Maree's scoring rate is 4.2 runs
per over.



4 A water pump fills a tank in 8 hours.
The tank contains 2 500 litres when full.
At what rate is the pump working?
Answer in L/h.

It pumps 2 500 L in 8 hours.
Divide: by 8
The rate is: 312.5 L/h

5 Jane delivers 690 bottles of milk in a 3 hour period.
What is her rate per hour?

She delivers 690 bottles in 3 hours.
Divide: by 3
The rate is: 130 bottles per hour

6 A motorist bought 40 litres of petrol. It cost \$51.60.
What was the cost of petrol per litre?

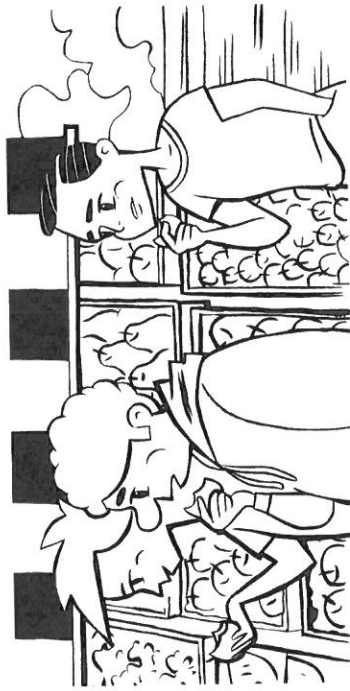
The petrol cost \$51.60 for 40 litres.
Divide: by 40
The rate is: \$1.29 / L



Calculating rates 1B

7 Calculate the rate of pay per hour.

- A** \$95 for 10 hours \$9.50/h.
B \$162 for 9 hours \$18/h.
C \$105 for 7 hours \$15/h.
D \$180 for 15 hours \$12/h.
E \$278.40 for 20 hours \$13.92/h.



8 Calculate the cost per kilogram.

- A** Oranges cost \$3.98 for 2 kg \$1.99 per kg
B Apples cost \$7.95 for 3 kg \$2.65 per kg
C Bananas cost \$7.70 for 2 kg \$3.85 per kg
D Tomatoes cost \$10.35 for 3 kg \$3.45 per kg
E Potatoes cost \$11.75 for 5 kg \$2.35 per kg

RATES

UNIT 1
TOPIC 1

9 Michaela scored 870 runs in 10 innings of cricket.
What was her average scoring rate of runs per innings?

87 runs per innings.

10 A tiler lays 56 square metres of floor tiles in 7 hours.
What is his average rate per hour?

8 m²/h.

11 The tide rose 42 cm in 6 hours.
What was the average rate of the tide rising?

7 cm/h.

12 The local football team kicked 168 goals in 28 matches.
What is their average scoring rate of goals per match?

6 goals per match.

13 A bricklayer lays 460 bricks in 5 hours.

This is an average rate of
92 bricks per hour.

14 A tree grows 156 cm
in 3 years.

This is an average growth
rate of 52 cm per year.



Calculating rates 2A

1

Calculate these rates.

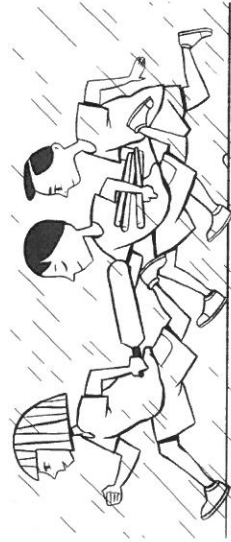
- A** 25 litres in 5 minutes is a rate of 5 L/min.
B 150 kilometres in 3 hours is a rate of 50 km/h.
C 100 metres in 10 seconds is a rate of 10 m/s.
D \$81 in 3 hours is a rate of \$ 27 per hour.
E \$630 in 5 days is a rate of \$ 126 per day.
F \$840 in 14 weeks is a rate of \$ 60 per week.
G 15.6 cm in 6 weeks is a rate of 2.6 cm/week.
H \$345 for 3 nights' accommodation is a rate of \$ 115 per night.

- I** \$37.50 for 3 metres of material is a rate of \$ 12.50 per metre.

- J** 400 litres in 16 hours is a rate of 25 L/h.

- K** 50 runs for 4 wickets is a rate of 12.5 runs per wicket.

- L** 144 mm of rain in 6 hours is a rate of 24 mm/h.



RATES

UNIT 1
TOPIC 1

2

Select the units that may be used to express the following rates by writing A to F in each box.

- A** typing rate **B** cost of potatoes
C pulse rate **D** population growth
E speed of a car **F** growth of a plant

- D** persons per year **F** millimetres per week
C beats per minute **A** words per minute
B dollars per kilogram **E** kilometres per hour

3

During a netball season, a team used four goalers. The table shows the number of goals each player scored.

Player's name	Goals scored	Number of attempts	Success rate
Pat	19	25	0.76
Helen	32	40	0.8
Malene	21	30	0.7
Peta	6	12	0.5

A Calculate the success rate for each player. (Goals scored ÷ number of attempts).

B Which player had the highest success rate? Helen.

Calculating rates 2B

RATES

UNIT 1
TOPIC 1

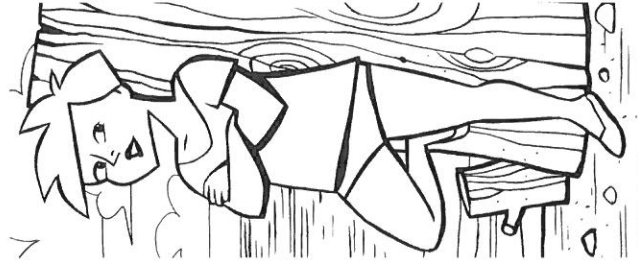
- 4** Shaun is paid \$340.80 for working 30 hours.
What is his hourly rate of pay?
(Answer to the nearest cent.)
\$11.36/h

- 5** A farmer produces 24 000 kg of potatoes
on a farm measuring 20 hectares.
What is the rate of production in kg/ha?
1200kg per hectare

- 6** A water pump fills a 1.5 kilolitre tank in 8 hours.
At what rate is a pump working?
(Answer in litres per hour.)
187.5L/h

- 7** A tree grows 312 mm in height
in 13 weeks.
Calculate its growth rate
in mm/week.
24 mm/week

- 8** Lisa's heart beats 396 times
in 6 minutes.
Calculate her heartbeat rate.
66 beats/min



The *density* of a substance can be expressed in grams per cubic centimetre (g/cm^3).
It is calculated by dividing the mass by its volume.

- 9** Calculate the density of a piece of metal
with mass 110 g and volume 20 cm^3 .
5.5g/cm³
- 10** A piece of wood has mass 450 g and volume 500 cm^3 .
Calculate its density.
0.9g/cm³
- 11** A piece of gold has a volume of 2.5 cm^3 .
Its mass is 30 grams. Calculate its density.
12g/cm³

The petrol consumption of a car is usually expressed in litres used per 100 km (L/100 km).

- 12** Calculate the petrol consumption
in L/100 km of a car that uses:
- | | | |
|----------|--------------------|-------------------|
| A | 20 L for 200 km. | <u>10L/100km</u> |
| B | 17 L for 200 km. | <u>8.5L/100km</u> |
| C | 27 L for 300 km. | <u>9L/100km</u> |
| D | 42 L for 500 km. | <u>8.4L/100km</u> |
| E | 60.8 L for 800 km. | <u>7.6L/100km</u> |